

**MARK SCHEME for the October/November 2010 question paper
for the guidance of teachers**

9702 PHYSICS

9702/41

Paper 4 (A2 Structured Questions), maximum raw mark 100

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

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Section A

- 1 (a) force per unit mass (*ratio idea essential*) B1 [1]
- (b) graph: correct curvature
from ($R, 1.0 g_S$) & at least one other correct point M1
A1 [2]
- (c) (i) fields of Earth and Moon are in opposite directions M1
either resultant field found by subtraction of the field strength
or any other sensible comment A1
so there is a point where it is zero A0 [2]
(allow $F_E = -F_M$ for 2 marks)
- (ii) $GM_E / x^2 = GM_M / (D - x)^2$ C1
(6.0×10^{24}) / (7.4×10^{22}) = $x^2 / (60R_E - x)^2$ C1
 $x = 54 R_E$ A1 [3]
- (iii) graph: $g = 0$ at least $\frac{2}{3}$ distance to Moon B1
 g_E and g_M in opposite directions M1
correct curvature (by eye) and $g_E > g_M$ at surface A1 [3]
- 2 (a) (i) no forces (of attraction or repulsion) between atoms / molecules / particles B1 [1]
- (ii) sum of kinetic and potential energy of atoms / molecules M1
due to random motion A1 [2]
- (iii) (random) kinetic energy increases with temperature M1
no potential energy
(so increase in temperature increases internal energy) A1 [2]
- (b) (i) zero A1 [1]
- (ii) work done = $p\Delta V$ C1
= $4.0 \times 10^5 \times 6 \times 10^{-4}$
= 240 J (*ignore any sign*) A1 [2]
- (iii)
- | change | work done / J | heating / J | increase in internal energy / J |
|--------|---------------|-------------|---------------------------------|
| P → Q | +240 | -600 | -360 |
| Q → R | 0 | +720 | +720 |
| R → P | -840 | +480 | -360 |
- (*correct signs essential*)
(*each horizontal line correct, 1 mark – max 3*) B3 [3]

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- 3 (a) (i) resonance B1 [1]
- (ii) amplitude 16 mm and frequency 4.6 Hz A1 [1]
- (b) (i) $a = (-)\omega^2 x$ and $\omega = 2\pi f$
 $a = 4\pi^2 \times 4.6^2 \times 16 \times 10^{-3}$
 $= 13.4 \text{ ms}^{-2}$ C1
C1
A1 [3]
- (ii) $F = ma$
 $= 150 \times 10^{-3} \times 13.4$
 $= 2.0 \text{ N}$ C1
A1 [2]
- (c) line always 'below' given line and never zero M1
peak is at 4.6 Hz (or slightly less) and flatter A1 [2]
- 4 (a) charge / potential (difference) (*ratio must be clear*) B1 [1]
- (b) (i) $V = Q / 4\pi\epsilon_0 r$ B1 [1]
- (ii) $C = Q / V = 4\pi\epsilon_0 r$ and $4\pi\epsilon_0$ is constant
so $C \propto r$ M1
A0 [1]
- (c) (i) $r = C / 4\pi\epsilon_0$
 $r = (6.8 \times 10^{-12}) / (4\pi \times 8.85 \times 10^{-12})$
 $= 6.1 \times 10^{-2} \text{ m}$ C1
C1
A1 [3]
- (ii) $Q = CV = 6.8 \times 10^{-12} \times 220$
 $= 1.5 \times 10^{-9} \text{ C}$ A1 [1]
- (d) (i) $V = Q/C = (1.5 \times 10^{-9}) / (18 \times 10^{-12})$
 $= 83 \text{ V}$ A1 [1]
- (ii) *either* energy = $\frac{1}{2}CV^2$
 $\Delta E = \frac{1}{2} \times 6.8 \times 10^{-12} \times 220^2 - \frac{1}{2} \times 18 \times 10^{-12} \times 83^2$
 $= 1.65 \times 10^{-7} - 6.2 \times 10^{-8}$
 $= 1.03 \times 10^{-7} \text{ J}$
or energy = $\frac{1}{2}QV$
 $\Delta E = \frac{1}{2} \times 1.5 \times 10^{-9} \times 220 - \frac{1}{2} \times 1.5 \times 10^{-9} \times 83$
 $= 1.03 \times 10^{-7} \text{ J}$ C1
C1
A1 [3]
(C1)
(C1)
(A1)

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- 5 (a) field into (the plane of) the paper B1 [1]
- (b) force due to magnetic field provides the centripetal force B1
 $mv^2 / r = Bqv$ C1
 $B = (20 \times 1.66 \times 10^{-27} \times 1.40 \times 10^5) / (1.6 \times 10^{-19} \times 6.4 \times 10^{-2})$ B1
 $= 0.454 \text{ T}$ A0 [3]
- (c) (i) semicircle with diameter greater than 12.8 cm B1 [1]
- (ii) new flux density = $\frac{22}{20} \times 0.454$ C1
 $B = 0.499 \text{ T}$ A1 [2]
- 6 (a) (i) e.g. prevent flux losses / improve flux linkage B1 [1]
- (ii) flux in core is changing B1
e.m.f. / current (induced) in core B1
induced current in core causes heating B1 [3]
- (b) (i) that value of the direct current producing same (mean) power / heating M1
in a resistor A1 [2]
- (ii) power in primary = power in secondary M1
 $V_P I_P = V_S I_S$ A1 [2]
- 7 (a) (i) e.g. electron / particle diffraction B1 [1]
- (ii) e.g. photoelectric effect B1 [1]
- (b) (i) 6 A1 [1]
- (ii) change in energy = $4.57 \times 10^{-19} \text{ J}$
 $\lambda = hc / E$ C1
 $= (6.63 \times 10^{-34} \times 3.0 \times 10^8) / (4.57 \times 10^{-19})$
 $= 4.4 \times 10^{-7} \text{ m}$ A1 [2]
- 8 (a) splitting of a heavy nucleus (*not atom/nuclide*) M1
into two (lighter) nuclei of approximately same mass A1 [2]
- (b) ${}^1_0\text{n}$
 ${}^4_2\text{He}$ (*allow* ${}^4_2\alpha$) M2
 ${}^7_3\text{Li}$ A1 [3]
- (c) emitted particles have kinetic energy B1
range of particles in the control rods is short / particles stopped in rods /
lose kinetic energy in rods B1
kinetic energy of particles converted to thermal energy B1 [3]

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Section B

- 9 (a) (i) non-inverting (amplifier) B1 [1]
- (ii) $(G =) 1 + R_2 / R_1$ B1 [1]
- (b) (i) gain = $1 + 100 / 820$ C1
output = 17 mV A1 [2]
- (ii) 9V A1 [1]
(R_2 / R_1 scores 0 in (a)(ii) but possible 1 mark in each of (b)(i) and (b)(ii)
($1 + R_1 / R_2$) scores 0 in (a)(ii), no mark in (b)(i), possible 1 mark in (b)(ii)
($1 - R_2 / R_1$) or R_1 / R_2 scores 0 in (a)(ii), (b)(i) and (b)(ii))
- 10 (a) (i) density $\times$ speed of wave (in the medium) B1 [1]
- (ii) $\rho = (7.0 \times 10^6) / 4100$
 $= 1700 \text{ kg m}^{-3}$ A1 [1]
- (b) (i) $I = I_T + I_R$ B1 [1]
- (ii) 1. $\alpha = (0.1 \times 10^6)^2 / (3.1 \times 10^6)^2$ C1
 $= 0.001$ A1 [2]
2. $\alpha \approx 1$ A1 [1]
- (c) *either* very little transmission at an air-skin boundary M1
(almost) complete transmission at a gel-skin boundary M1
when wave travels in or out of the body A1 [3]
or no gel, majority reflection (M1)
with gel, little reflection (M1)
when wave travels in or out of the body (A1)
- 11 (a) (i) unwanted random power / signal / energy B1 [1]
- (ii) loss of (signal) power / energy B1 [1]
- (b) (i) *either* signal-to-noise ratio at mic. $= 10 \lg (P_2 / P_1)$ C1
 $= 10 \lg \{ \{2.9 \times 10^{-6}\} / \{3.4 \times 10^{-9}\} \}$
 $= 29 \text{ dB}$ A1
maximum length $= (29 - 24) / 12$ C1
 $= 0.42 \text{ km} = 420 \text{ m}$ A1 [4]
- or* signal-to-noise ratio at receiver $= 10 \lg (P_2 / P_1)$ (C1)
at receiver, $24 = 10 \lg (P / \{3.4 \times 10^{-9}\})$
 $P = 8.54 \times 10^{-7} \text{ W}$ (A1)
power loss in cables $= 10 \lg \{ \{2.9 \times 10^{-6}\} / \{8.54 \times 10^{-7}\} \}$ (C1)
 $= 5.3 \text{ dB}$
length $= 5.3 / 12 \text{ km}$
 $= 440 \text{ m}$ (A1)

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- (ii) use an amplifier
coupled to the microphone
(*repeater amplifiers scores no mark*)
- M1
A1 [2]

- 12 (a) (carrier wave) transmitted from Earth to satellite (1)
satellite receives greatly attenuated signal (1)
signal amplified and transmitted back to Earth
at a different (carrier) frequency B1
different frequencies prevent swamping of uplink signal B1
e.g. of frequencies used (6/4 GHz, 14/11 GHz, 30/20 GHz) (1)
(*two B1 marks plus any two other for additional physics*) (1)
- B2 [4]

- (b) advantage: e.g. much shorter time delay M1
because orbits are much lower A1
e.g. whole Earth may be covered (M1)
in several orbits / with network (A1)
- disadvantage: e.g. *either* must be tracked M1
or limited use in any one orbit A1
more satellites required for continuous operation [4]